

Feature Screening in Breast Cancer Diagnosis: A Visual Exploration of the Wisconsin Diagnostic Dataset

DSM050 – Data Visualisation, Midterm Coursework

[Word count: 1,978 words]

1. Research Context and Problem Definition

Breast cancer is the most commonly diagnosed cancer among women worldwide. The World Health Organisation estimates 2.4 million new diagnoses and 694,000 deaths globally in 2024, and it is the leading cancer in women in 164 of 186 countries surveyed (World Health Organisation, 2024). For decades, cytological assessment of fine-needle aspirate (FNA) biopsies, examining the size, shape, and texture of cell nuclei under a microscope, has been a standard low-cost diagnostic tool. Digitising this process opened the door to quantitative, computational approaches. Street, Wolberg and Mangasarian (1993) showed that nuclear characteristics extracted from digitised FNA images (radius, texture, perimeter, concavity, and related measures) could be computed automatically and used to classify tumours as benign or malignant, producing the Breast Cancer Wisconsin (Diagnostic) dataset used in this report. It remains a benchmark for machine-learning-based diagnosis research, including recent classification studies (Yedjou et al., 2021).

Rather than building a predictive classifier, this report focuses on **exploratory data analysis** using visual techniques to understand which of the 30 measured tumour characteristics carry diagnostic signal, and how that signal is structured across related groups of features.

Research objective: to identify, through univariate and multivariate visual analysis, which tumour measurements most reliably distinguish malignant from benign diagnoses, and whether the dataset’s “worst” (most extreme) measurements carry more diagnostic signal than its “mean” measurements.

Target audience: clinical and bioinformatics researchers evaluating which FNA-derived measurements warrant closer attention when screening tumour samples. Findings are exploratory and descriptive, not diagnostic or predictive, and are not intended to inform individual patient care.

2. Research Questions

RQ1: Which tumour characteristics most strongly separate malignant from benign cases?

With 30 candidate features, an analyst cannot inspect every measurement with equal attention. Prior work using this dataset suggests a small subset of features, particularly those describing nuclear shape irregularity (concavity, concave points) and size (radius, perimeter, area), carry most of the classification signal (Yedjou et al., 2021). The first research question tests this directly and visually, using effect size to rank all 30 features by how well they separate the two diagnostic classes, rather than relying on classification accuracy alone, which can obscure which variables are doing the discriminating.

RQ2: Do the “worst” (largest/most extreme) measurements discriminate malignancy better than the “mean” measurements? The dataset’s structure, each of 10 base characteristics recorded as a mean, a standard error, and a “worst” value (Street, Wolberg & Mangasarian, 1993), is a design decision worth investigating. If malignancy is associated with irregular, heterogeneous cell populations, the most extreme cells in a sample may be more diagnostically informative than the sample average. The second research question evaluates this using correlation strength, directly comparing each mean/worst feature pair,

with practical implications for which measurements should be prioritised in feature engineering or manual review.

Both questions are feasible within the dataset’s scope (569 labelled samples, complete records, established provenance) and are addressed using univariate and multivariate visualisation rather than predictive modelling.

3. Dataset and Preprocessing

Dataset. The Breast Cancer Wisconsin (Diagnostic) dataset, accessed via `sklearn.datasets.load_breast_cancer`, contains 569 tumour samples, each described by 30 numeric features and a binary diagnosis (357 benign, 212 malignant, a 63%/37% split). The 30 features are 10 base nuclear characteristics (radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension), each expressed three ways: **mean**, **standard error**, and **worst** (the largest/most extreme value observed across cells in the sample) (Street, Wolberg & Mangasarian, 1993). The exact CSV used is included in `data/`.

Handling missing values. An explicit check (`df.isnull().sum()`) confirmed zero missing values and zero duplicate rows across all 569 records. No imputation, row removal, or placeholder substitution was therefore required.

Managing dimensionality. Thirty numeric features are too many to visualise meaningfully at once. Two preprocessing decisions were used to manage this issue:

1. Features were grouped by measurement type (`mean_*`, `*_error`, `worst_*`), so visualisations compare feature families rather than 30 unrelated columns. This grouping is what makes RQ2 answerable.
2. A **Cohen’s d** effect size (Cohen, 1988) was computed for every feature against diagnosis, giving one comparable ranking of discriminative power.

Scaling. A z-score standardisation (`StandardScaler`) was fitted across all 30 features, because the raw features each have very different scales, **mean area** reaches into the thousands of square pixels, while **mean smoothness** is a fraction between 0 and 0.2, so any comparison of magnitude or spread across features would therefore be dominated by whichever variable has the largest raw units.

4. Exploratory Data Analysis (EDA)

4.1 Univariate analysis (RQ1)

Class balance. The diagnosis variable (nominal, two levels) splits 357 benign to 212 malignant (Figure 1), since the benign group both establishes the “typical” distribution and contains more samples.

Distribution shape. Before comparing features by class, six representative `mean_*` features were plotted as simple histograms across all 569 tumours (Figure 2), to check overall shape and skew ahead of any group-based comparison.

Ranked discriminative power. Every one of the 30 features was scored with Cohen’s d against the diagnosis label; Figure 3 shows the top 12, ranked by magnitude.

Distribution by diagnosis. Figure 4 returns to the six top-ranked features from Figure 3, now split by diagnosis, to show the actual separation in each distribution that the effect-size ranking is summarising numerically.

4.2 Multivariate analysis (RQ2)

Mean vs. worst, feature by feature. Figure 5 compares, for each of the 10 base characteristics, the absolute correlation with diagnosis for its `mean_*` version against its `worst_*` version.

Relationship between mean and worst measurements. Figure 6 plots mean radius against worst radius directly, coloured by diagnosis, to show how the two versions of the same measurement relate to one another and to class.

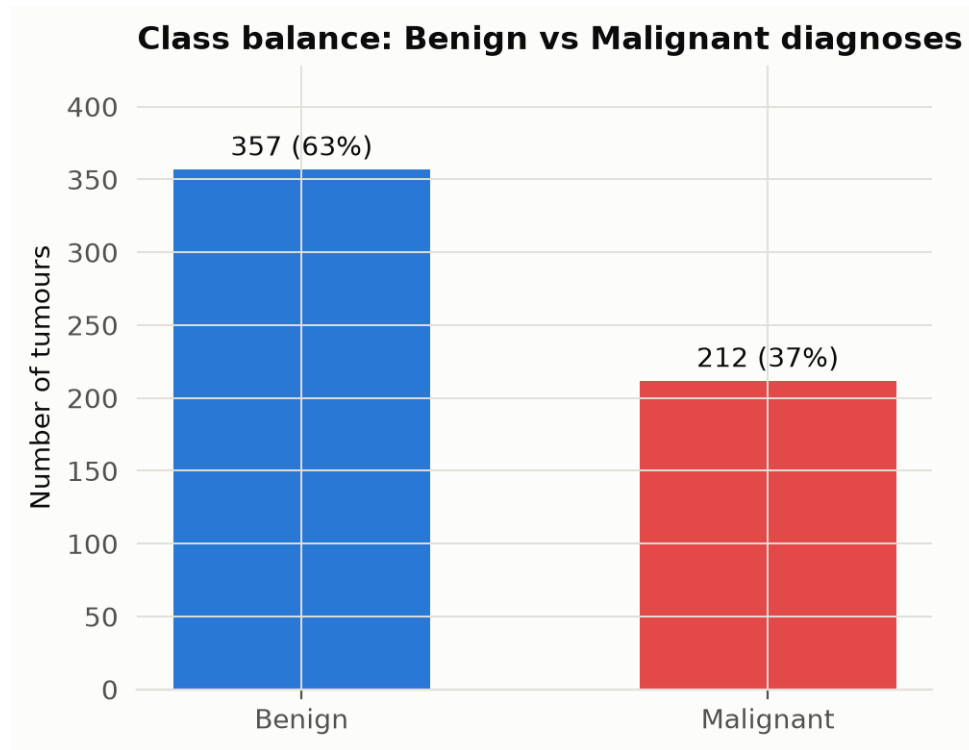


Figure 1: Class balance between benign and malignant diagnoses.

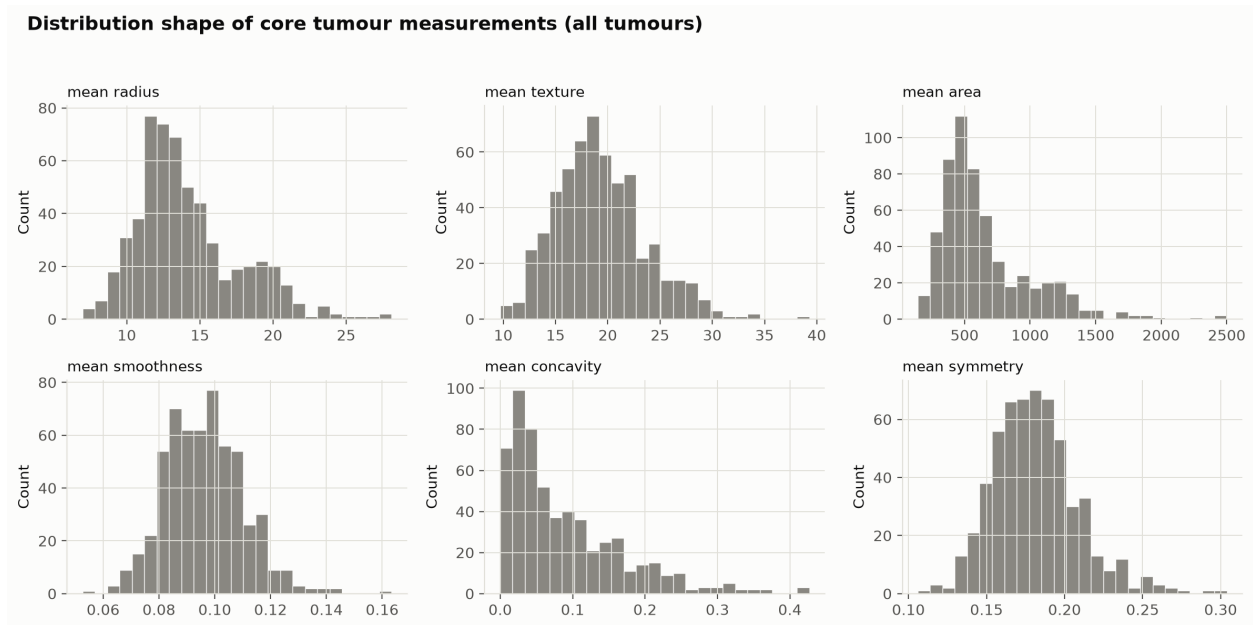


Figure 2: Distribution shape of six core tumour measurements, all tumours combined.

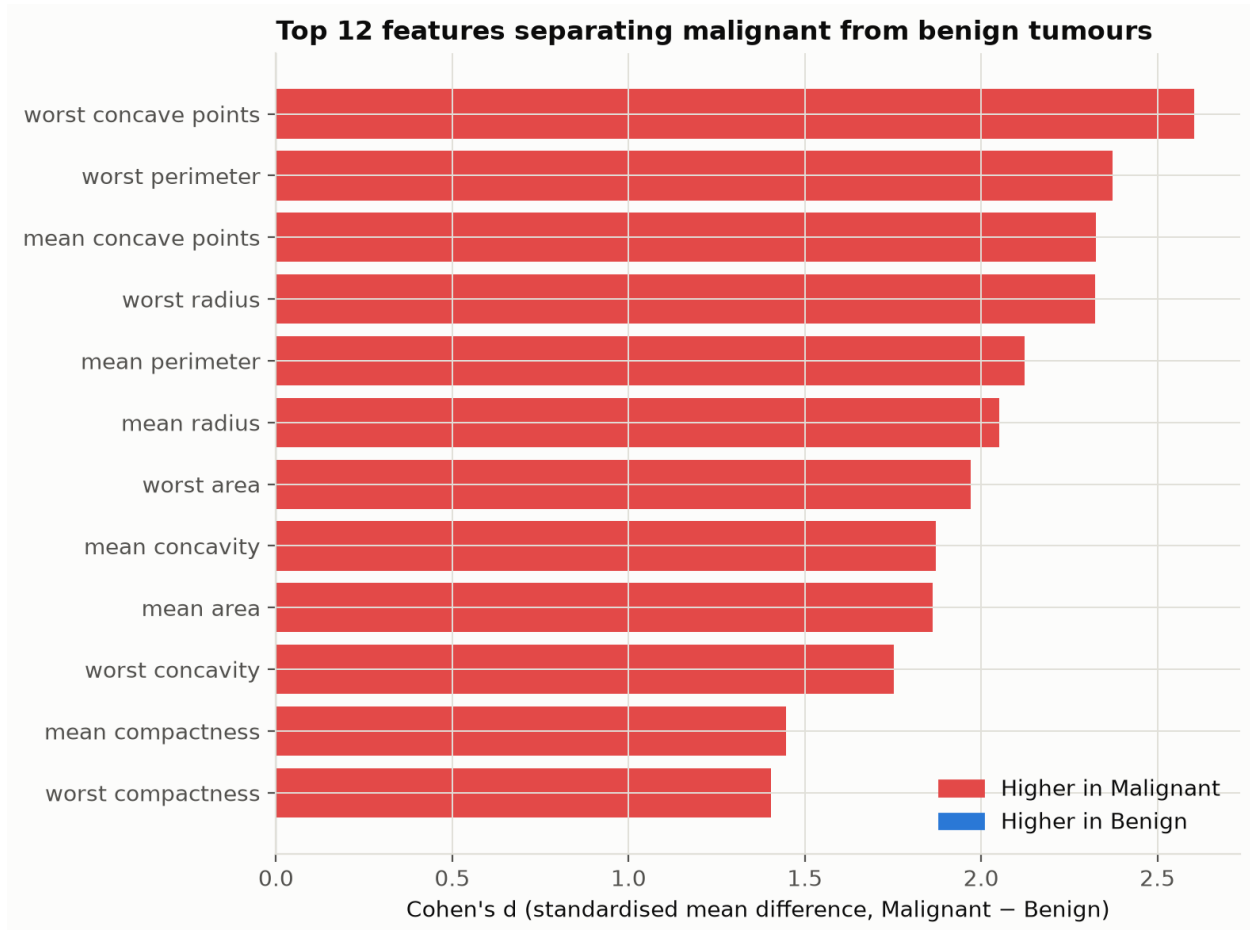


Figure 3: Top 12 features ranked by standardised mean difference (Cohen's d) between malignant and benign tumours.

Distribution by diagnosis: top discriminating features

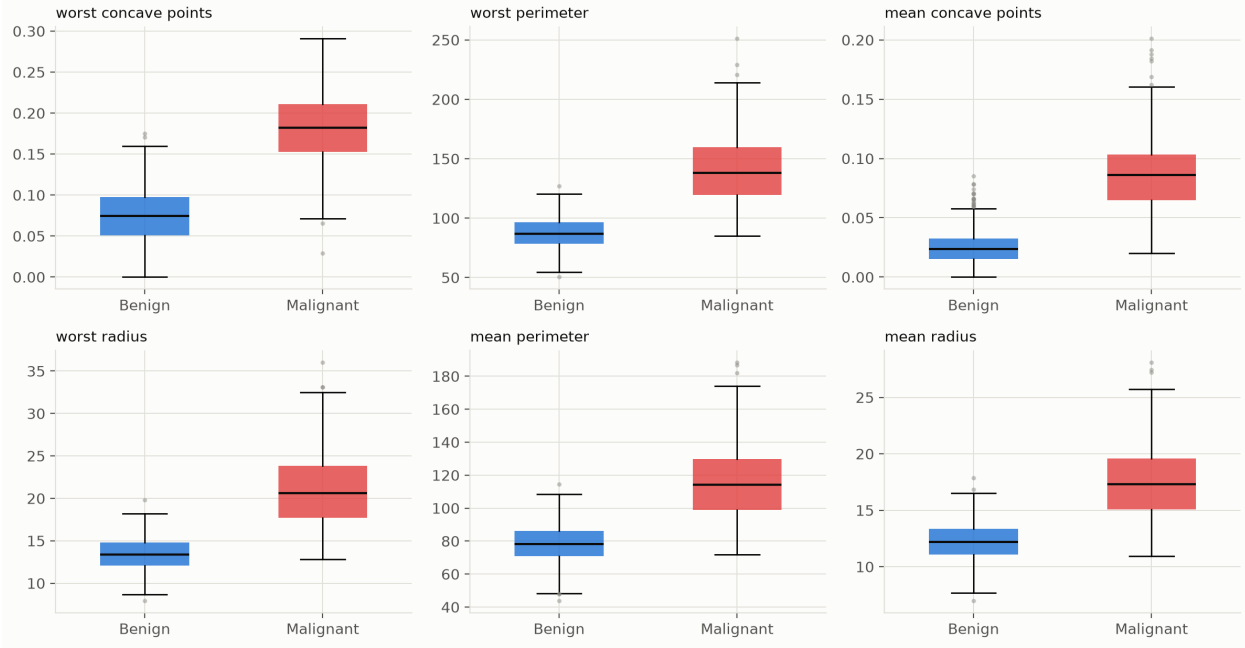


Figure 4: Boxplot distributions of the six most discriminating features, split by diagnosis.

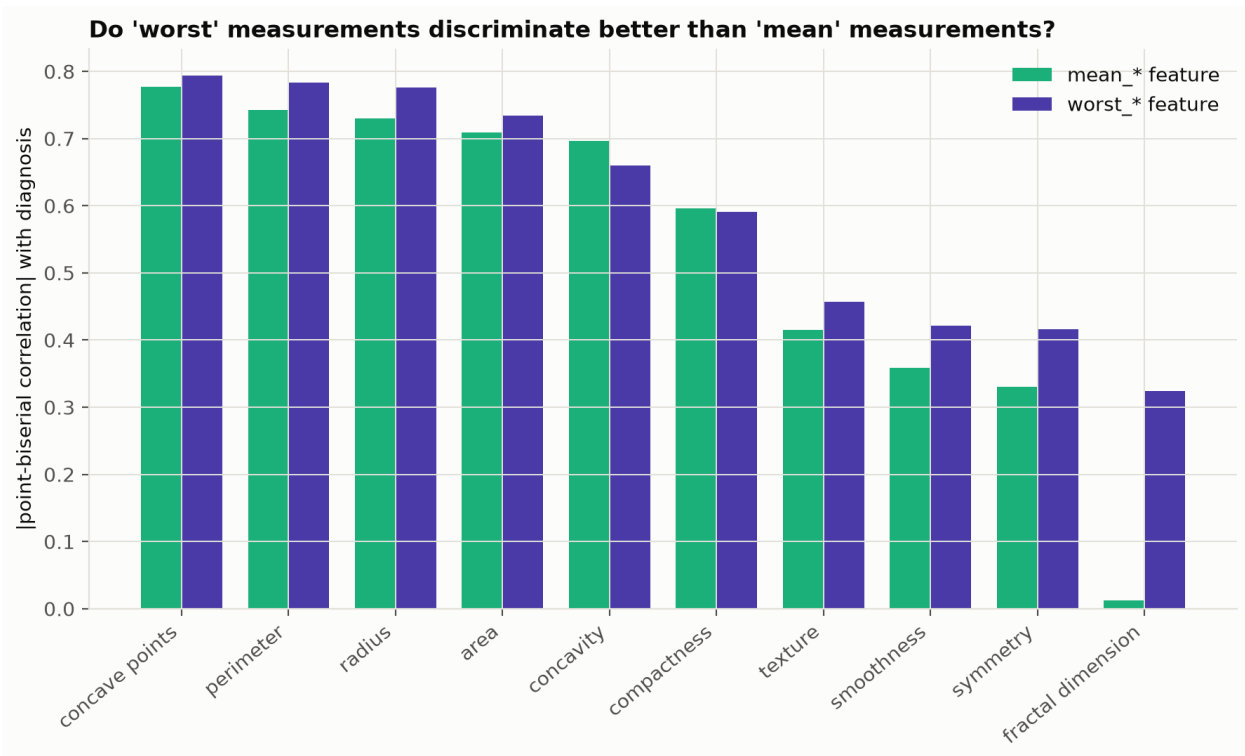


Figure 5: Grouped comparison of mean_* vs worst_* |correlation| with diagnosis, for the four base characteristics with the strongest signal.

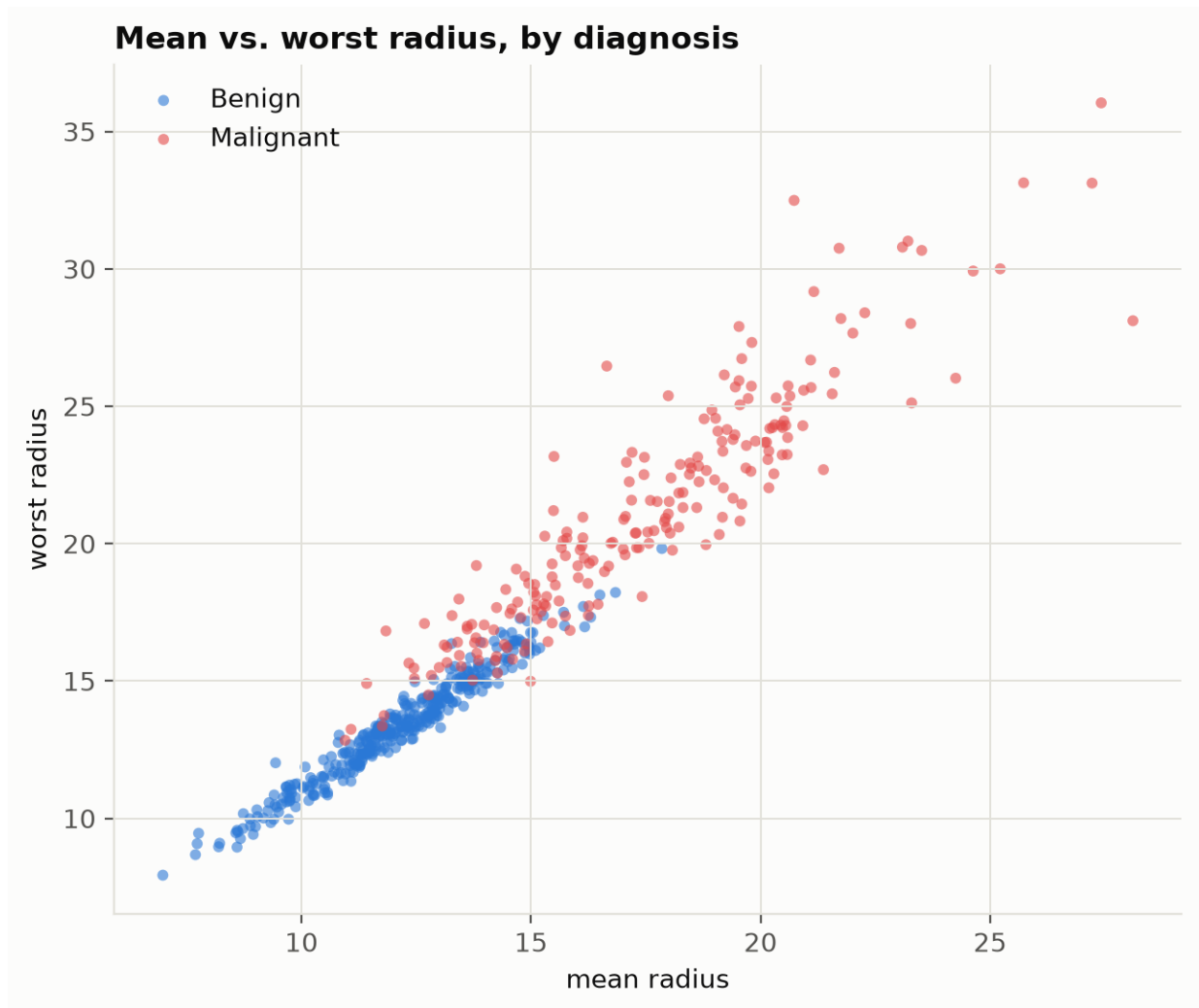


Figure 6: Mean vs. worst radius, coloured by diagnosis.

Internal correlation structure. Figure 7 places the correlation matrix of the `mean_*` feature block alongside that of the `worst_*` feature block, to check whether the two families of features are structured similarly or differently among themselves.

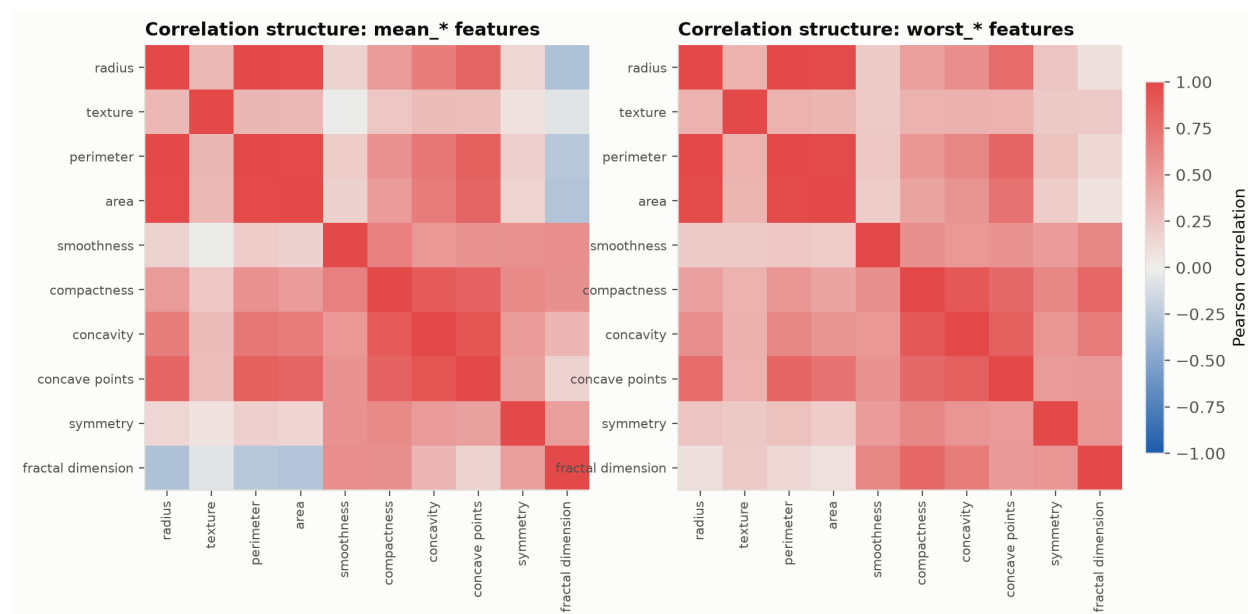


Figure 7: Correlation structure within the `mean_*` feature block (left) and the `worst_*` feature block (right).

Two-dimensional class separation. Finally, Figure 8 plots the two most discriminating `worst_*` features (identified in Figure 3) against each other, to see how well class separation holds when two strong features are considered jointly rather than one at a time.

4.3 Dashboard

Figure 9 consolidates the findings above into a single, static, at-a-glance view designed for a clinical/bioinformatics research audience who need a summary rather than the full sequence of exploratory charts. For example, a researcher deciding which measurements to prioritise when manually reviewing a batch of borderline samples. The layout follows a deliberate information hierarchy (stated directly on the figure): dataset scope, class balance, which features matter most, the mean-vs-worst framing from RQ2, and finally a visual demonstration of class separation on the two strongest features. Colour is used consistently with every other figure in this report (blue = benign, red = malignant; a fixed categorical assignment, never reused for anything else), so the dashboard reads as part of the same visual system rather than a separate, disconnected artefact.

5. Interpretation and Discussion of Findings

5.1 RQ1: Which features separate malignant from benign tumours?

The top-ranked features by Cohen’s d , `worst_concave_points` ($d=2.60$), `worst_perimeter` (2.37), `mean_concave_points` (2.33), `worst_radius` (2.32), `mean_perimeter` (2.12), and `mean_radius` (2.05), all share a clear physical theme, they describe nuclear size (radius, perimeter) and boundary irregularity (concave points, the count of inward dents in a nucleus’s outline). This indicates that malignant nuclei in this dataset tend to be both larger and more irregularly shaped than benign nuclei, consistent with established cytological understanding of malignancy. By contrast, the lowest-ranked features, `mean_fractal_dimension`, `texture_error`, `symmetry_error` (all with $|d|$ near 0), are either noise-sensitive error terms or a highly abstract complexity measure, and appear to carry little of the signal that actually distinguishes the two classes. The boxplots in Figure 4 support this ranking visually, the top features show a real shift in

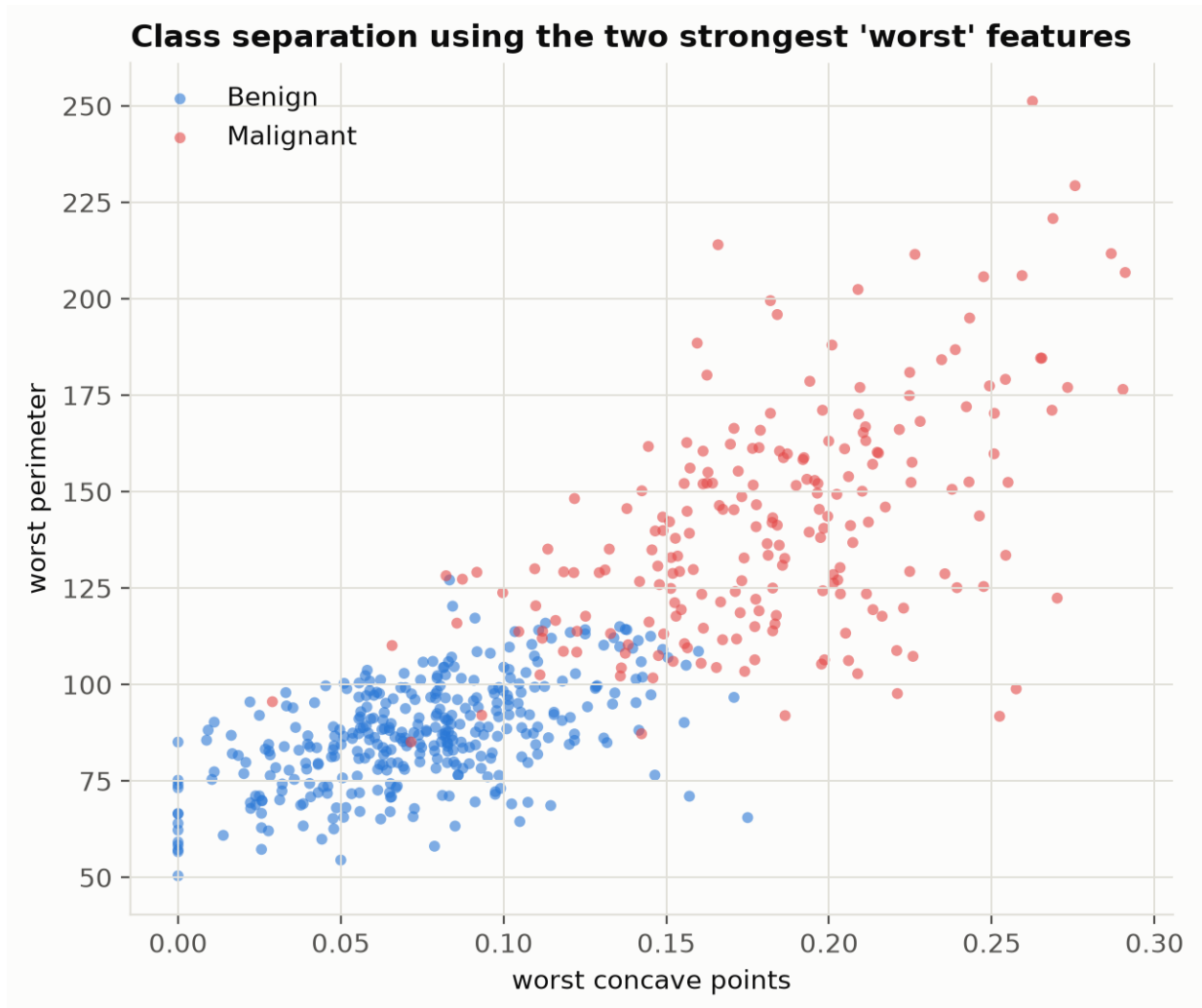


Figure 8: Class separation using the two strongest `worst_*` features (numeric $\times$ numeric $\times$ categorical).

Breast Cancer Diagnostic Dataset — Feature Screening Dashboard

Audience: clinical researchers evaluating which tumour measurements warrant closest review

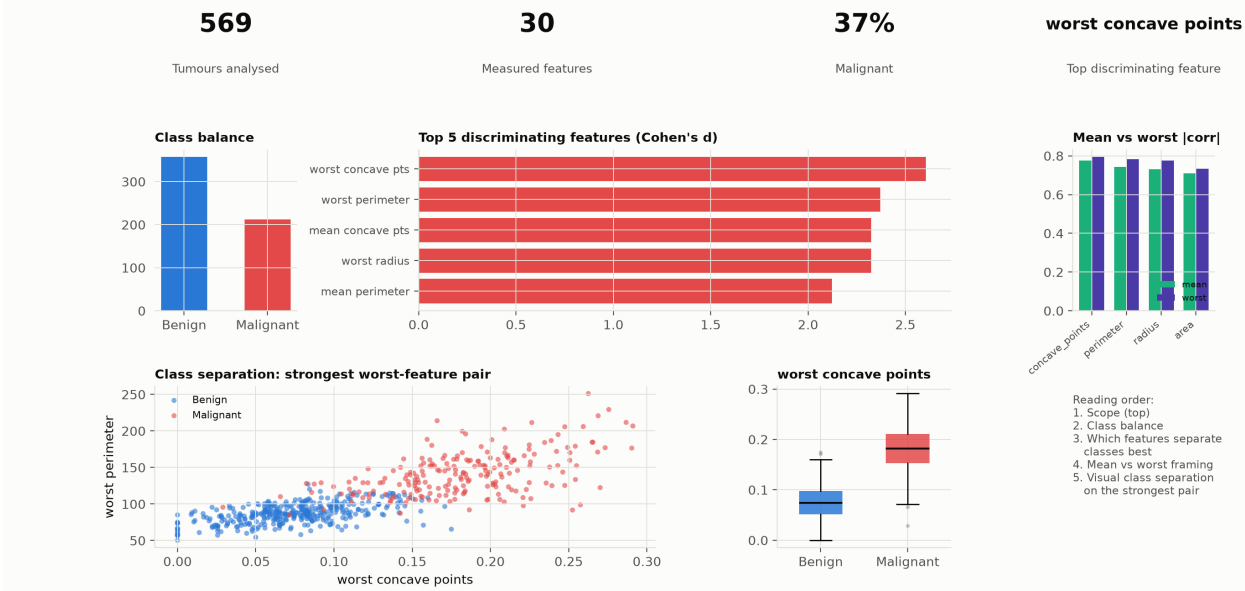


Figure 9: Feature-screening dashboard summarising class balance, top discriminating features, the mean-vs-worst comparison, and class separation on the strongest feature pair.

distribution between classes, though with some overlap rather than a clean distinction, meaning these measurements are strongly informative in the aggregate but not individually decisive for every case. For a clinician or researcher, this suggests size and shape-irregularity measurements warrant closer attention when screening samples, rather than symmetry or fractal-dimension-based metrics.

5.2 RQ2: Do “worst” measurements discriminate better than “mean” measurements?

The mean-vs-worst comparison (Figure 5) shows the pattern is not uniform across features. For most characteristics, concave points, perimeter, radius, area, the worst* *version correlates more strongly with diagnosis than the mean** version, most dramatically for fractal_dimension, where the mean carries almost no signal ($r=0.013$) but the worst value shows a moderate relationship ($r=0.324$). This supports the idea that the single most extreme cell in a sample can reveal diagnostic signal that gets diluted when averaged across an entire sample. However, concavity and compactness break this pattern, with the mean correlating marginally more strongly than the worst value, suggesting that for these two characteristics, the overall shape of the cell population is more informative than any single outlier cell. Figures 6 and 7 support this, the mean and worst versions of radius are highly correlated with each other, and the internal correlation structure of the two feature blocks is broadly similar, indicating that “worst” measurements are not capturing an entirely different signal, but rather sharpening the same underlying signal for size/shape features specifically. This implies that dataset design should not default to reporting either mean or worst values alone, but consider which is more informative on a per-feature basis.

5.3 Implications, limitations, and audience relevance

For the stated audience of clinical and bioinformatics researchers, these findings suggest that a shortlist of nuclear size and irregularity measurements, rather than all 30 features, could reasonably guide manual review of borderline samples. However, several limitations are present in this conclusion. This analysis is descriptive and exploratory. Effect size and correlation quantify separation and linear association but do not establish predictive accuracy, causal mechanism, or clinical validity, which would require a proper

classification model with held-out validation. The dataset itself is a single-institution, historical sample (Street, Wolberg & Mangasarian, 1993), and its class balance (63% benign, 37% malignant) may not reflect the true relationship in a screening population, meaning the “typical” distribution shown here is shaped by this particular sample rather than the wider population. Finally, both effect size and Pearson-style correlation summarise mean-level, broadly linear separation, and may understate the diagnostic value of features with more complex, non-linear relationships to malignancy. A natural follow-up would be predictive modelling with proper validation, to test whether the features identified here as most discriminative also perform best in an actual classifier.

6. Declaration of AI Tool Use

Generative AI (Claude, Anthropic) was used during this coursework as follows: **(1)** drafting and debugging the Python/Matplotlib preprocessing and visualisation code in `notebook/DSM050_breast_cancer_analysis.ipynb`, including resolving library version issues; **(2)** applying a structured colour and layout methodology (consistent categorical colour assignment, no dual-axis charts, accessible colour pairing) across the nine figures and dashboard; **(3)** searching for and verifying the citations in Section 7; **(4)** drafting the structural skeleton and prose for Sections 1–4.

AI was **not** used to generate, alter, or fabricate data, to select which statistical findings to report, or to write Section 5 (Interpretation and Discussion). All AI-assisted code was executed and its output independently reviewed; all AI-suggested citations were cross-checked against their original sources before use.

7. Reference List

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GitHub Repo Link: <https://github.com/jayminic/dsm050-midterm-breast-cancer-visualisation>